1. Concept of Confidence Intervals

A **Confidence Interval (CI)** is a **range of values** that likely contains the **true population parameter** (e.g., mean, proportion) with a specified **level of confidence**.

It is expressed as:

$$\text{Confidence Interval} = \text{Point Estimate} \pm \text{Margin of Error}$$

where:

- **Point Estimate**: The sample statistic (e.g., sample mean, sample proportion).
- **Margin of Error (ME)**: The uncertainty in the estimate, based on sample size and variability.

Common Confidence Levels

- **90% Confidence Interval** → $\alpha = 0.10$ (Z = 1.645)
- **95% Confidence Interval** → $\alpha = 0.05$ (Z = 1.96) **(Most common)**
- **99% Confidence Interval** → $\alpha = 0.01$ (Z = 2.576)

Formula for Confidence Interval for Mean

For a population mean (μ), if the population standard deviation (σ) is known:

$$CI = \bar{X} \pm Z \times \frac{\sigma}{\sqrt{n}}$$

If σ **is unknown**, use the sample standard deviation (s) and replace Z with the **t-score** from the t-distribution:

$$CI = \bar{X} \pm t \times \frac{s}{\sqrt{n}}$$

where:

- $\bar{X}$ = sample mean
- Z = Z-score (from standard normal distribution)
- t = t-score (from t-table for small samples)
- σ or s = standard deviation
- n = sample size

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2. Numerical Examples

Example 1: Confidence Interval for Population Mean (σ Known)

Problem:
A sample of **n = 100** students has an average test score of $\bar{X} = 75$ with a population standard deviation $\sigma = 10$. Find the **95% confidence interval** for the population mean.

Solution:

- Given: $\bar{X} = 75$, $\sigma = 10$, $n = 100$, $Z = 1.96$ (for 95% confidence)
- Standard error:

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{10}{\sqrt{100}} = \frac{10}{10} = 1$$

- Margin of Error:

$$ME = Z \times SE = 1.96 \times 1 = 1.96$$

- Confidence Interval:

$$CI = 75 \pm 1.96$$

$$CI = (73.04, 76.96)$$

Interpretation: We are **95% confident** that the true population mean lies between **73.04 and 76.96**.

Example 2: Confidence Interval for Population Mean (σ Unknown, Small Sample)

Problem:
A sample of **n = 16** students has an average GPA of **3.2**, with a **sample standard deviation** $s = 0.4$. Find the **99% confidence interval** for the population mean.

Solution:

- Given: $\bar{X} = 3.2$, $s = 0.4$, $n = 16$, $t = 2.947$ (from t-table for 99% CI, df = 15)
- Standard error:

$$SE = \frac{s}{\sqrt{n}} = \frac{0.4}{\sqrt{16}} = \frac{0.4}{4} = 0.1$$

- Margin of Error:

$$ME = t \times SE = 2.947 \times 0.1 = 0.2947$$

- Confidence Interval:

$$CI = 3.2 \pm 0.2947$$

$$CI = (2.91, 3.49)$$

Interpretation: We are **99% confident** that the true population mean GPA lies between **2.91 and 3.49**.

Example 3: Confidence Interval for Population Proportion

Problem:
A survey of **n = 500** voters found that **p̂ = 60% (0.60) support a candidate**. Find the **95% confidence interval** for the true proportion of voters who support the candidate.

Solution:

- Given: $\hat{p} = 0.60$, $n = 500$, $Z = 1.96$ (for 95% CI)
- Standard error:

$$SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.60(0.40)}{500}}$$

$$SE = \sqrt{\frac{0.24}{500}} = \sqrt{0.00048} = 0.0219$$

- Margin of Error:

$$ME = Z \times SE = 1.96 \times 0.0219 = 0.0429$$

- Confidence Interval:

$$CI = 0.60 \pm 0.0429$$

$$CI = (0.557, 0.643)$$

Interpretation: We are **95% confident** that the true proportion of voters who support the candidate is between **55.7% and 64.3%**.

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3. Key Takeaways

Concept	Formula	Use Case
CI for Mean (σ known)	$CI = \bar{X} \pm Z \times \frac{\sigma}{\sqrt{n}}$	Large samples, known population standard deviation
CI for Mean (σ unknown)	$CI = \bar{X} \pm t \times \frac{s}{\sqrt{n}}$	Small samples, unknown population standard deviation
CI for Proportion	$CI = \hat{p} \pm Z \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	Estimating population proportions

4. Interpretation of Confidence Intervals

1. A **95% confidence level** means that if we repeated the study **100 times**, about **95 times**, the true population parameter would fall within our CI.
2. A **wider interval** means **higher confidence** but **lower precision**.
3. **Larger sample sizes** reduce the margin of error and provide **narrower, more precise confidence intervals**.
4. **If a CI for a mean or proportion includes a certain value (e.g., 0 for effect sizes), we cannot conclude statistical significance.**